

Arboretum Eco-Quest

Solve outdoor ecology checkpoints

At a glance: Read the living collection and solve the checkpoints before time runs out. Plan about 80 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: tree identification, ecosystems, observation, classification evidence (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- 4 sets of clue cards, one card per checkpoint (plus 1 spare set; card set in this guide)
- 1 laminated dichotomous key per team, keyed to the route (plus 1 spare; key printed in this guide)
- 4 route maps with the checkpoints marked (plus 2 spares; route map in this guide)
- 4 clipboards (plus 1 spare)
- 4 evidence token bags (plus 1 spare), with evidence tokens for each checkpoint (about 30 tokens; cut-apart token sheet in this guide)
- pencils, shared
- optional QR codes that open the Arboretum Explorer record at each checkpoint

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

- 01 Set up one station per team with the materials above, sorted and ready.
- 02 Stage the scoring sheet and any reference values before testing begins.
- 03 No PPE: this is an outdoor observation walk with no chemicals, water, soil, or instruments. Set up sun and hydration support, run the weather and heat-index check, confirm the buddy system and route boundaries, and confirm the approved route with the arboretum host.
- 04 Load the activity slides and ready the projected timer.

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Read the living collection and solve the checkpoints before time runs out." Take a quick prediction by show of hands.

Phase 2 Predict and plan the route

0:05 - 0:12

Each team takes a quick prediction about which clue will be most reliable, then numbers the checkpoints on the route map in an order that avoids backtracking. No walking yet.

Phase 3 Constraints

0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Walk and solve

0:18 - 0:44

Teams walk their route, use the clue cards and the dichotomous key to answer each checkpoint, record the justifying clue, and collect the evidence token. Circulate, ask which clue led to each answer, and resist naming the tree for them.

Phase 5 Record evidence and collect tokens

At each checkpoint, teams write the one specific clue that justifies their answer and take the evidence token.

Phase 6 Check answers

Before returning, each team confirms every answer cites a real observed clue, not a guess, and has a token for each solved checkpoint.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, and reset the station. Collect the route packs, the keys, the clipboards, and the evidence tokens; there is no PPE to remove and nothing to wash.

TROUBLESHOOTING

Problem: A team backtracks and wastes time. **Fix:** Have them re-number the remaining checkpoints into a no-backtrack order; treat it as a route-planning coaching moment, since pace is scored.

Problem: A team is stuck at a checkpoint and cannot reach a name. **Fix:** Send them back to the first either-or choice on the key, have them point to the actual leaf or bark feature that decides it, and use the optional QR code only to confirm after they commit to an answer.

Problem: A team answers from memory instead of the clues. **Fix:** Require them to write the specific feature (leaf shape, bark, branching, or seed) for each answer; an answer with no cited clue scores zero on evidence quality.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to observation as evidence.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Which clue was most reliable for telling trees apart?
- Where did a key send you wrong, and why?
- What makes a good piece of field evidence?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Checkpoint accuracy	50
Evidence quality	20
Pace	20
Teamwork and conduct	10
Total	100

CLEANUP AND SOURCES

Collect the route maps, clue cards, dichotomous keys, clipboards, and evidence tokens and return them to the bin, and collect score slips. Nothing is consumed and there are no surfaces to wipe.

SOURCES Ambler Arboretum self-guided tour (<https://ambler.temple.edu/arboretum/our-gardens/self-guided-tour>); Arboretum Explorer interactive collection map (<https://ambler.arboretumexplorer.org/>)

Checkpoint clue cards

Print ONE set per team (4) plus 1 spare set, then cut apart. Each card names the arboretum area; teams walk there, identify the tagged tree with the key, and record the one clue that decided it. Staff: confirm the tagged tree at each post before the run. Laminating is optional; teams write on these, so plain cardstock is fine.

Checkpoint 1

token ○

Pinetum

south of the Learning Center, along Loop Drive

Identify the tagged tree here. Your ID:

Deciding clue (circle one): leaf · bark · branching
· seed

The specific feature you saw: _____

Checkpoint 2

token ○

Oak Canopy

southeast of the Research and Collaboration Building

Identify the tagged tree here. Your ID:

Deciding clue (circle one): leaf · bark · branching
· seed

The specific feature you saw: _____

Checkpoint 3

token ○

Beech Grove

west of Dixon Hall, across the bluestone Accessible Path

Identify the tagged tree here. Your ID:

Deciding clue (circle one): leaf · bark · branching
· seed

The specific feature you saw: _____

Checkpoint 4

token ○

Native Formal Garden

southeast of Dixon Hall; the black gum allee

Identify the tagged tree here. Your ID:

Deciding clue (circle one): leaf · bark · branching
· seed

The specific feature you saw: _____

Checkpoint 5

token ○

Trident maple

a Delaware Valley tree-of-record; staff mark its post

Identify the tagged tree here. Your ID:

Deciding clue (circle one): leaf · bark · branching
· seed

The specific feature you saw: _____

Checkpoint 6

token ○

Katsura

a Delaware Valley tree-of-record; staff mark its post

Identify the tagged tree here. Your ID:

Deciding clue (circle one): leaf · bark · branching
· seed

The specific feature you saw: _____

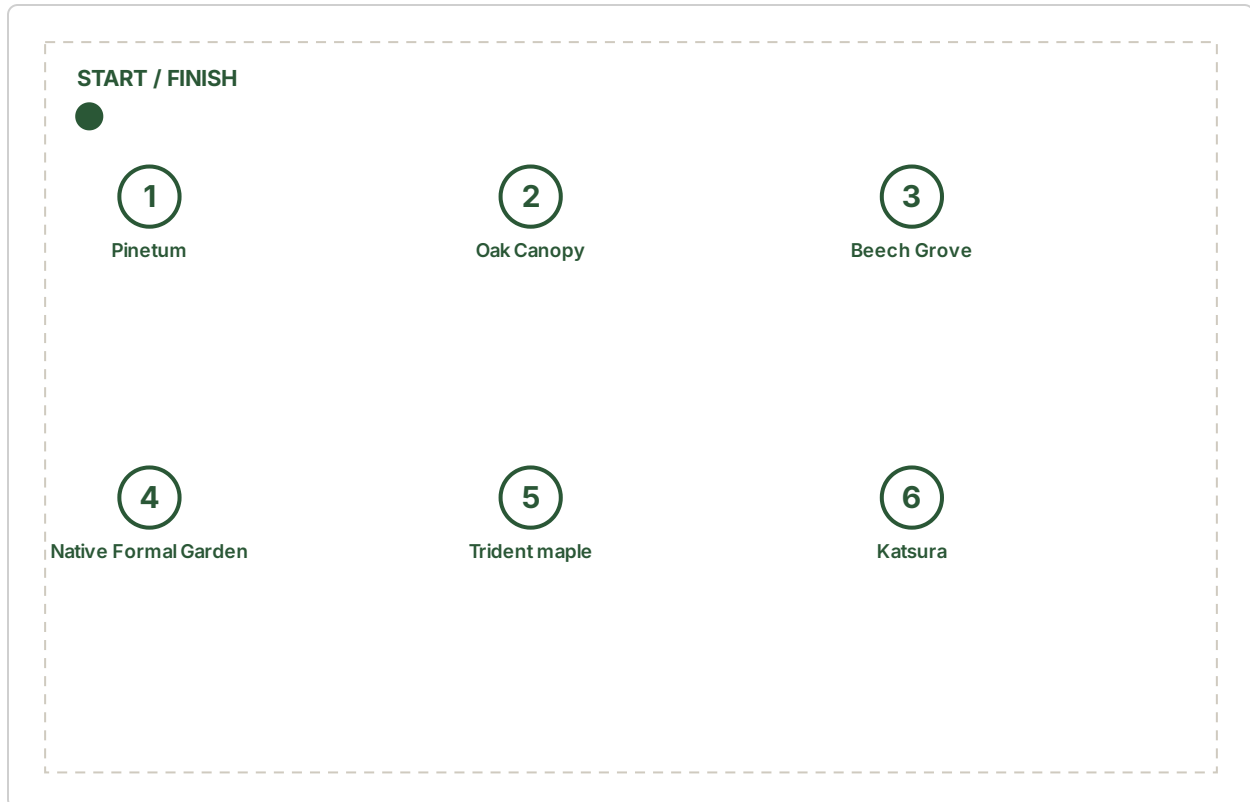
Tree dichotomous key

Keyed to the six trees on this route. Print 1 per team (4) plus 1 spare and laminate for reuse. Start at step 1 and follow the choice that matches what you see. Staff: if you swap a checkpoint tree, update the matching couplet.

STEP	WHAT TO LOOK FOR	THEN
1a	Leaves are needles in soft bundles (evergreen)	Eastern white pine
1b	Leaves are broad and flat	go to 2
2a	Leaves grow in opposite pairs	go to 3
2b	Leaves grow alternately (staggered)	go to 4
3a	Leaf is one blade with three palm-like lobes	Trident maple
3b	Leaf is a single blade, heart-shaped, edge smooth	Katsura
4a	Leaf edge is lobed; acorns on or under the tree	Oak
4b	Leaf edge is smooth or finely toothed	go to 5
5a	Bark smooth gray, buds long and pointed	American beech
5b	Leaf glossy, bark dark and blocky, red fall color	Black gum

Route map

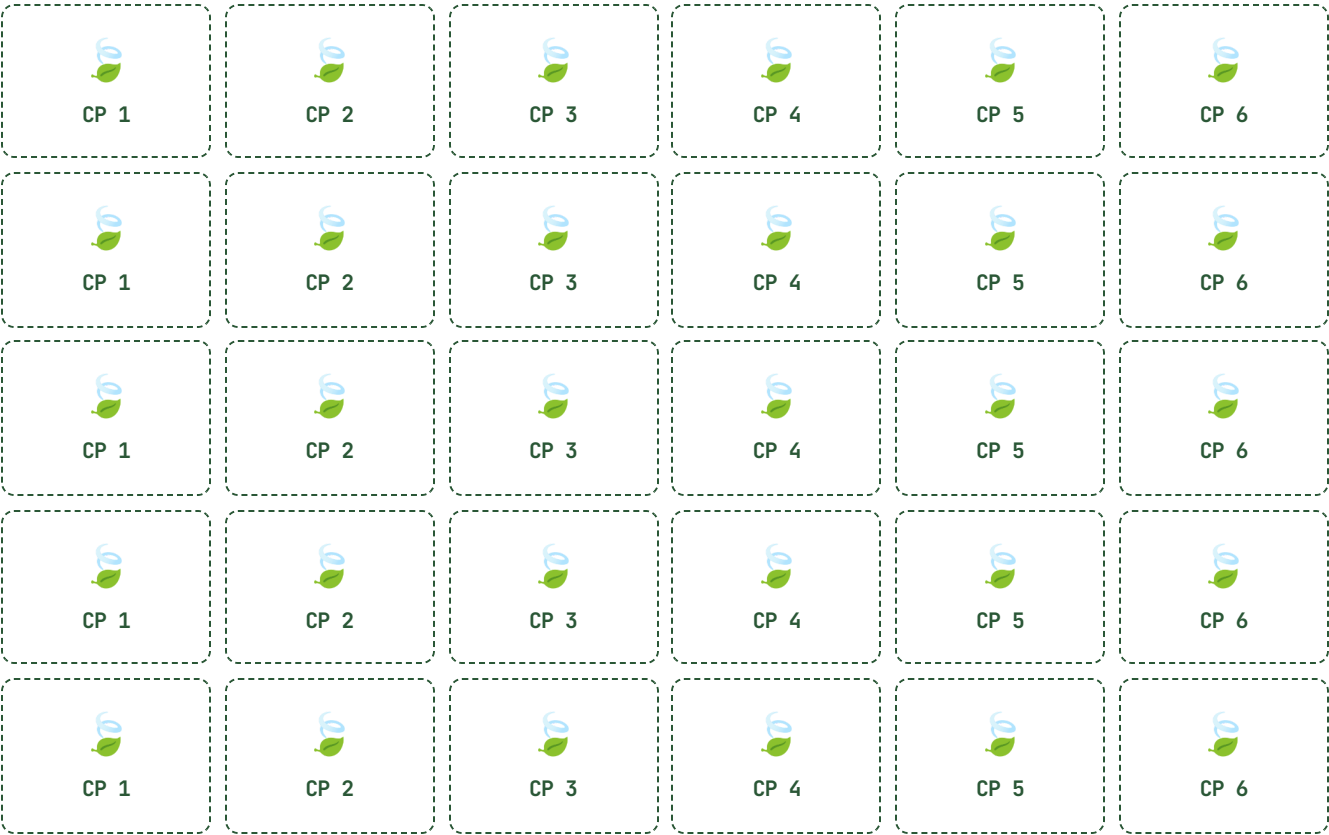
Six checkpoints through the Ambler Arboretum collection areas. Walk them in an order that avoids backtracking. Staff: the layout below is a schematic; place a numbered post at each area and confirm the walking path and boundaries with the arboretum host before the run. Print 4 (plus 2 spares). Base map: Ambler Arboretum self-guided tour and Arboretum Explorer.



- 01 **1. Pinetum** — south of the Learning Center, along Loop Drive
- 02 **2. Oak Canopy** — southeast of the Research and Collaboration Building
- 03 **3. Beech Grove** — west of Dixon Hall, across the bluestone Accessible Path
- 04 **4. Native Formal Garden** — southeast of Dixon Hall; the black gum alley
- 05 **5. Trident maple** — a Delaware Valley tree-of-record; staff mark its post
- 06 **6. Katsura** — a Delaware Valley tree-of-record; staff mark its post

Evidence tokens (cut apart)

Cut apart. Each team collects one token per solved checkpoint. This sheet makes 30 tokens (4 teams + 1 spare set of 6). Print on cardstock; no lamination needed.



Instructor answer key (do not hand out)

Keep this with the instructor. The expected tree at each checkpoint and the clue that decides it. If a tagged specimen has changed, update the checkpoint and the matching couplet in the key.

CP	AREA	EXPECTED TREE	DECIDING CLUE
1	Pinetum	Eastern white pine	needles in soft bundles of five
2	Oak Canopy	Oak (Bender oak)	alternate leaf with rounded or pointed lobes; acorns
3	Beech Grove	American beech	smooth gray bark and long, pointed buds
4	Native Formal Garden	Black gum	alternate glossy leaves, dark blocky bark, brilliant red fall color
5	Trident maple	Trident maple	opposite leaves with three palm-like lobes
6	Katsura	Katsura	opposite heart-shaped leaves with a smooth edge